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(54) **STRAWBERRY PLANT NAMED ‘MALIBU’**

(50) Latin Name: *Fragaria x ananassa*
Varietal Denomination: **Malibu (a.k.a. ‘109221’)**

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patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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(51) **Int. Cl.**
A01H 5/08 (2018.01)

(52) **U.S. Cl.**
USPC **Plt./209**

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See application file for complete search history.

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(57) **ABSTRACT**

The present invention provides a new and distinct straw-
berry plant designated as ‘Malibu’ (a.k.a. ‘109221’).

3 Drawing Sheets

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Latin name of the genus and species: *Fragaria x anan-*
assa.
Varietal denomination: ‘Malibu’ (a.k.a. ‘109221’).

BACKGROUND OF THE INVENTION

The present invention relates to a new and distinct straw-
berry plant designated as ‘Malibu’ (a.k.a. ‘109221’).
‘Malibu’ is a short-day strawberry plant.

‘Malibu’ (a.k.a. ‘109221’) is the result of a controlled-
cross between a female parent cultivar designated 105429
(strawberry plant named ‘Dani’, U.S. Plant Pat. No. 25,533)
and a male parent cultivar designated 105218 (strawberry
plant named ‘Zarina’, U.S. Plant Pat. No. 23,936) made by
the Inventor and was first fruited in Watsonville, Calif.
growing fields. Following selection and during testing, the
plant was originally designated ‘109221’ and subsequently
named ‘Malibu’.

This new strawberry plant was asexually reproduced via
runners (stolons) by the inventor at Watsonville, Calif.
Asexual propagules from the original source have been
tested in Watsonville growing fields and to a limited extent,
grower fields in high elevation. The properties of this new
plant were found to be transmissible by such asexual repro-
duction. This plant is stable and reproduce true to type in
successive generations of asexual reproduction.

BRIEF SUMMARY OF THE INVENTION

This invention relates to a new and distinctive strawberry
plant designated as ‘Malibu’. This strawberry plant is pri-
marily adapted to the climate and growing conditions of the
central coast of California. This region provides the neces-
sary temperatures required for it to produce a strong vigor-
ous plant and to remain in fruit production from March
through October. The nearby Pacific Ocean provides the
needed humidity and moderate day temperatures and even-
ing chilling to maintain fruit quality for the production
months.

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The following traits and photographs in combination
distinguish strawberry plant ‘Malibu’ from known straw-
berry varieties. In addition, this new strawberry plant was
confirmed to be a unique strawberry germplasm when tested
against the California Seed & Plant Lab, Inc. (Elverta,
Calif.) database using Short Sequence Repeats (SSRs).
Plants for the botanical measurements in the present appli-
cation were grown as annuals. Any color references are
made to The Royal Horticultural Society Colour Chart, 1995
Edition, except where general terms of ordinary dictionary
significance are used. The fruit produced by each new
cultivar is attractive and of excellent quality.

DESCRIPTION OF THE DRAWINGS

The accompanying color photographs depict various char-
acteristics of the cultivars as nearly true as possible to make
color reproductions.

FIG. 1 shows fruits of ‘Malibu’.

FIG. 2 shows sliced fruits of ‘Malibu’.

FIG. 3 shows ‘Malibu’ plants.

DETAILED DESCRIPTION OF THE INVENTION

‘Malibu’ (a.k.a. ‘109221’)

This invention relates to a new and distinctive short-day
type strawberry cultivar designated as ‘Malibu’. It is pri-
marily adapted to the climate and growing conditions of the
central coast of California. This region provides the neces-
sary temperatures required for it to produce a strong vigor-
ous plant and to remain in fruit production from March
through October. The nearby Pacific Ocean provides the
needed humidity and moderate day temperatures and even-
ing chilling to maintain fruit quality for the production
months.

The following traits in combination distinguish straw-
berry variety ‘Malibu’ from the known strawberry varieties.
Plants for the botanical measurements in the present appli-

cation were grown as annuals. In the following description, color references are made to The Royal Horticultural Society Colour Chart, 1995 Edition, except where general terms of ordinary dictionary significance are used.

The detailed botanical description in Table 1 was observed when the plants were 33 weeks after planting. The plants observed were grown in a farm field with an outdoor yield trial plot (without cover) on a polyethylene-film-covered-raised-bed. A grower standard 1.25 mil thickness of polyethylene film was used to cover raised bed. All plants were spaced 14 inches apart. The field having a very sandy soil was located on a commercial strawberry farm, which is near Watsonville, Calif.

‘Malibu’ has not been observed under all possible environmental conditions, and the phenotype may vary significantly with variations in environment. The following observations, measurements, and comparisons describe this plant as grown under normal conditions in Watsonville, Calif. unless otherwise noted.

TABLE 1

Malibu		
CharType	Characteristic	Malibu
General	Plant Habit	annual
	Plant Growth Habit	semi-upright
	Plant Height	31 cm
	Plant Width	37 cm
	Plant Width-Crown	10 cm
	Density of foliage, vigor	light
	Plant vigor	moderate
Leaf	Terminal leaflet width (mm)	90
	Terminal leaflet length (mm)	85
	No. teeth/terminal leaflet:	16
	Shape of the terminal leaflet base	acute
	Shape of terminal leaflet in cross-section	concave
	Margin description of the terminal	serrate to crenate
Limbs	Color of upper side of leaves	137A
	Color of lower side of leaves	137C
	Leaf blistering	weak
	Leaf glossiness	medium
	Petiole length (cm)	17
	Petiole diameter (mm)	3
	Petiole color	145B
	Petiolule length (mm)	10
	Petiolule diameter (mm)	3
	Attitude of hairs on petiole and pedicel	horizontal to upwards
	Stipule pubescence	sparse
	Stipule length (cm)	3.5
	Stipule size	medium
	Stipule width (cm)	1
Inflorescence	Stipule anthocyanin	present
	Stipule color (color code)	145A
	Pedicel color (color code)	145A
	Peduncle length (cm)	25
	Peduncle size	medium to large
	Peduncle attitude	erect
	Peduncle pubescence, attitude of hairs	medium, upwards
	Inflorescence position relative to foliage	above
	Flower arrangement of petals	touching
	Petal length (cm)	1.3
	Petal width (cm)	1.4
	Petal number per flower	5 to 6
	Upper Petal color	155A
	Lower Petal color	155B
	Calyx diameter (cm)	3.9
	Corolla diameter (cm)	2.6
	Sepal length (cm)	1.7
	Sepal width (cm)	0.9
	Time of flowering	March

TABLE 1-continued

Malibu		
CharType	Characteristic	Malibu
5	(50% of plants in bloom)	
	Shape of stigma	capitate
	Color of stigma	14A
	Length of style (mm)	1.5
	Color of style	7D
10	Color of the ovary	145B
	Length of the stamens (mm)	4.7
	Number of stamen	21 to 24
	Anther color	13B
	Shape of anther	dorsifixed
	Size of anther	medium
15	Amount of pollen	abundant
	Color of pollen	14A
	Color of filament	149D
	Length of filament (mm)	3 to 6
	Number of flowers per truss	3 to 5
Stolon	Stolon number	1
	Stolon anthocyanin	181A
20	Widest diameter of stolon	4.19
	At leaf attachment (mm)	
Fruit	Stolon color	145A
	Number of fruit per truss	3 to 5
	Fruit length (cm)	4.2
	Fruit width (cm)	4.3
25	Fruit skin color	45A
	Fruit flesh color excluding core	41A
	Fruit core length (cm)	4.1
	Fruit core width (cm)	1.7
	Fruit core color	37A
	Fruit weight (g)	24.5
30	Predominant fruit shape	conic to long conic
	Shape difference between primary & secondary fruits	Similar shape
	Width of band without of achenes	medium
	Fruit glossiness	strong
	Position of achenes	below surface
	Achene color	150B
35	Achenes per fruit	399
	Achene weight (g)	0.24
	Position of calyx	even to above
	level of adherence of calyx	medium
	Color of calyx	137A
	Firmness of flesh	medium
40	Evenness of flesh color	nearly even
	Sweetness (brix)	11
	pH	3.48
	Yield (g per plant per season)	2208

When ‘Malibu’ is compared to the proprietary female parent ‘Dani’ (U.S. Plant Pat. No. 25,533), the petiole of ‘Malibu’ leaves is thicker than that of ‘Dani’ leaves. ‘Malibu’ produces fruits earlier than ‘Dani’ in the spring season. The flowers of ‘Malibu’ are visible above the canopy, while the flowers of ‘Dani’ are mostly below the canopy.

When ‘Malibu’ is compared to the proprietary male parent ‘Zarina’ (U.S. Plant Pat. No. 23,936), ‘Malibu’ is a medium-large sized plant, but ‘Zarina’ is a small-medium sized plant. Fruits of ‘Malibu’ are uniformly smooth and conic, while fruits of ‘Zarina’ are long-conic.

When ‘Malibu’ is compared to the check variety ‘Monterey’ (U.S. Pat. No. 19,767), ‘Malibu’ does not produce runners during the fruit production period, while ‘Monterey’ produces stolons during the fruiting period. The fruit yield of ‘Monterey’ is significantly greater than ‘Malibu’. In addition, fruits of ‘Monterey’ are less uniform than the fruits of ‘Malibu’ that are uniformly conical shaped.

When ‘Malibu’ is compared to the check variety ‘Merced’ (U.S. Pat. No. 25,436), the leaf trifoliate of ‘Malibu’ differs

from that of ‘Merced’. ‘Malibu’ foliage allows for relatively more open canopy when compared to ‘Merced’ that has a closed canopy.

TABLE 2

Comparison of fruit features between ‘Malibu’ and the proprietary male andfemale parents				
HYBRID ID	HYBRID NAME	FRUIT WIDTH (mm)	FRUIT HEIGHT (mm)	FRUIT RATIO (Height/Width)
105218	Male Parent (Zarina)	41.05	48.19	1.17
105429	Female Parent (Dani)	41.99	48.86	1.16
109221	Malibu	38.04	44.62	1.17
HYBRID ID	HYBRID NAME	FRUIT SHAPE*	HARDNESS (newtons)	Yield (g/clone)
105218	Male Parent (Zarina)	7	7.52	974
105429	Female Parent (Dani)	6	7.60	2076
109221	Malibu	6	7.73	553

*Fruit shape: 1. Oblate; 2. Globose; 3. Fan Lobes; 4. Necked; 5. Short wedge; 6. Symmetric conic; 7. Conic; 8. Long conic; 9. Long wedge

TABLE 3

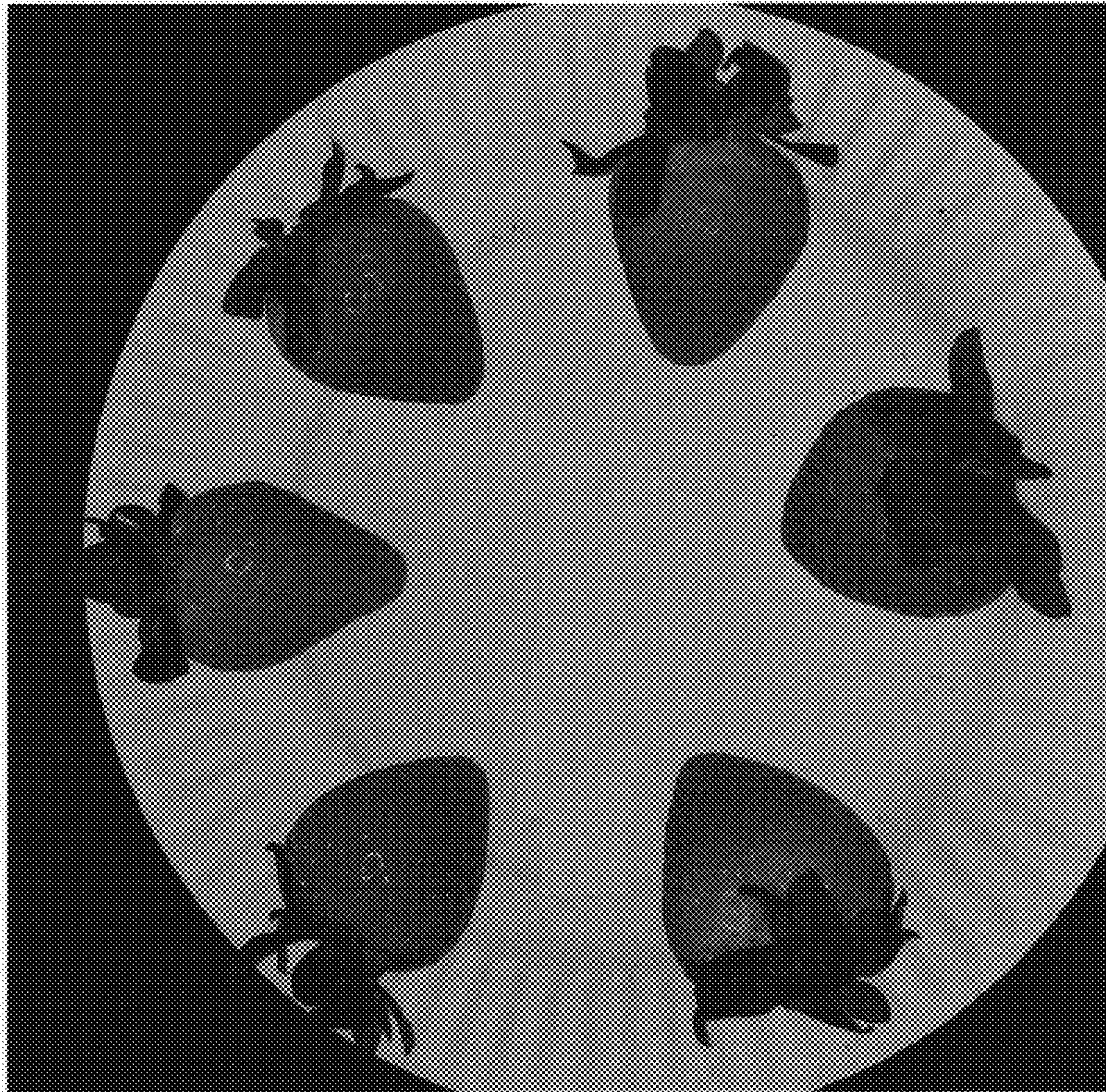
Comparison of fruit features between ‘Malibu’ and the check varieties				
HYBRID ID	HYBRID NAME	FRUIT WIDTH (mm)	FRUIT HEIGHT (mm)	FRUIT RATIO (Height/Width)
Check Variety	Monterey	43.70	48.33	1.11
Check Variety	Merced	40.40	45.19	1.12
109221	Malibu	36.96	43.38	1.17
HYBRID ID	HYBRID NAME	FRUIT SHAPE*	HARDNESS (newtons)	Yield (g/clone)
Check Variety	Monterey	6	9.04	840
Check Variety	Merced	7	8.81	949
109221	Malibu	7	7.62	589

*Fruit shape: 1. Oblate; 2. Globose; 3. Fan Lobes; 4. Necked; 5. Short wedge; 6. Symmetric conic; 7. Conic; 8. Long conic; 9. Long wedge

The invention claimed is:
1. A new and distinct cultivar of strawberry plant named ‘Malibu’ substantially as shown and described herein.

* * * * *

Figure 1



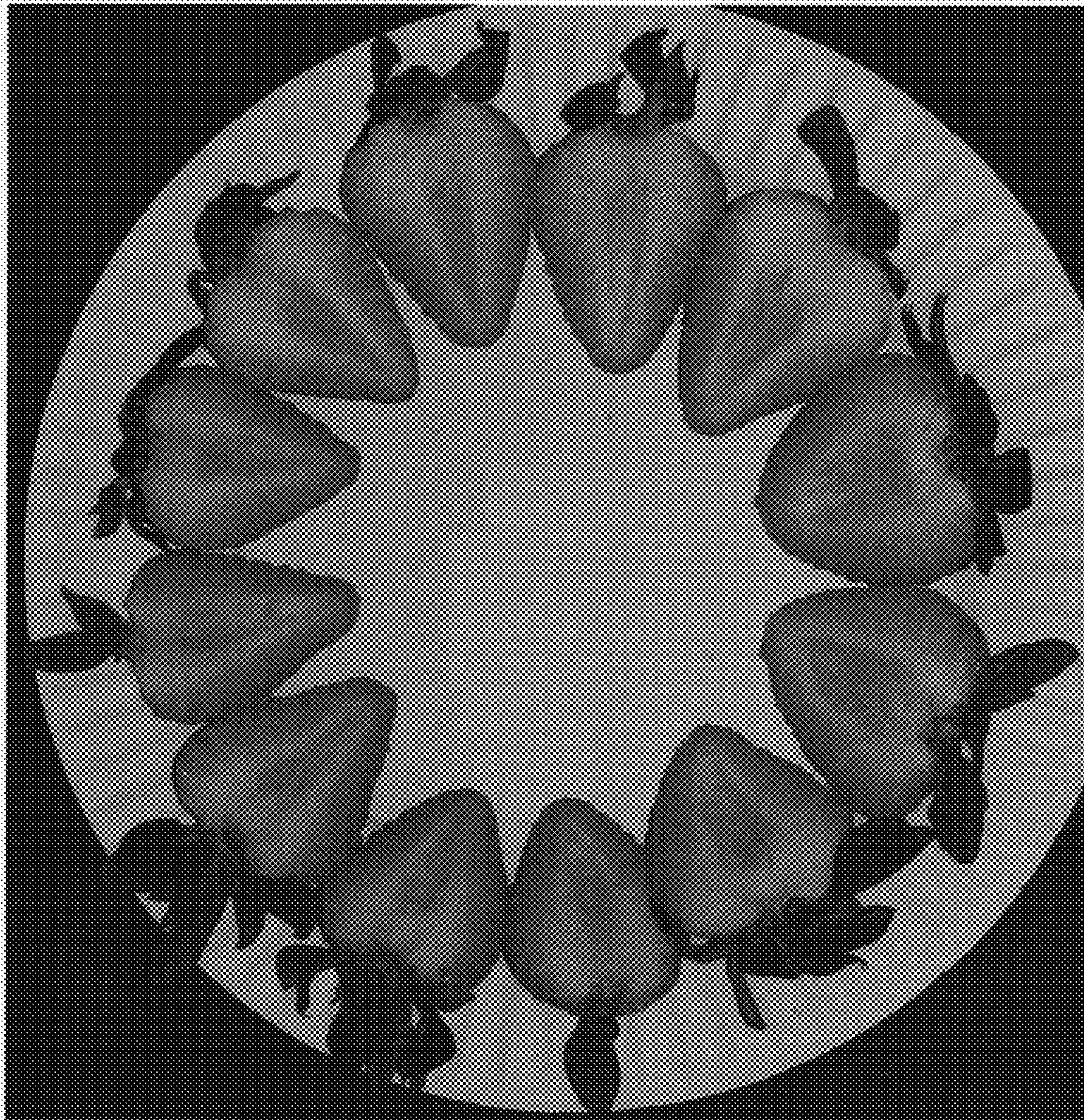


Figure 2



Figure 3